Self-Evaluation of Single Cycle Core.

```
# Test the RISC-V processor:
#   add, sub, and, or, slt, addi, lw, sw, beq, jal
# If successful, it should write the value 25 to address 100
#       RISC-V Assembly         Description                 Address   Machine Code
main:   addi x2, x0, 5          # x2 = 5                    0         00500113
        addi x3, x0, 12         # x3 = 12                   4         00C00193
        addi x7, x3, -9         # x7 = (12 - 9) = 3         8         FF718393
        or   x4, x7, x2         # x4 = (3 OR 5) = 7         C         0023E233
        and  x5, x3, x4         # x5 = (12 AND 7) = 4       10        0041F2B3
        add  x5, x5, x4         # x5 = 4 + 7 = 11           14        004282B3
        beq  x5, x7, end        # shouldn't be taken        18        02728863
        slt  x4, x3, x4         # x4 = (12 < 7) = 0         1C        0041A233
        beq  x4, x0, around     # should be taken           20        00020463
        addi x5, x0, 0          # shouldn't execute         24        00000293
around: slt  x4, x7, x2         # x4 = (3 < 5) = 1          28        0023A233
        add  x7, x4, x5         # x7 = (1 + 11) = 12        2C        005203B3
        sub  x7, x7, x2         # x7 = (12 - 5) = 7         30        402383B3
        sw   x7, 84(x3)         # [96] = 7                  34        0471AA23
        lw   x2, 96(x0)         # x2 = [96] = 7             38        06002103
        add  x9, x2, x5         # x9 = (7 + 11) = 18        3C        005104B3
        jal  x3, end            # jump to end, x3 = 0x44    40        008001EF
        addi x2, x0, 1          # shouldn't execute         44        00100113
end:    add  x2, x2, x9         # x2 = (7 + 18) = 25        48        00910133
        sw   x2, 0x20(x3)       # [100] = 25                4C        0221A023
done:   beq  x2, x2, done       # infinite loop             50        00210063
```

CODE:

00500113

00C00193

FF718393

0023E233

0041F2B3

004282B3

02728863

0041A233

00020463

00000293

0023A233

005203B3

402383B3

0471AA23

06002103

005104B3

008001EF

00100113

00910133

0221A023

00210063

Task Instructions for RISC-V Single Cycle Processor Verification

1. **Machine Code Loading**:
   Copy the provided machine code and load it into your processor's Instruction Memory.
2. **Processor Verification**:
   Verify the functionality of your RISC-V Single Cycle Processor by running the machine code according to the provided figure (reference diagram).
3. **Capture Screenshots**:
   During and after verification, take clear and properly labeled screenshots showing:
   - The loaded instruction memory.
   - The execution flow (key signal changes, register updates, etc.).
   - The final result verifying correct operation.
4. **Prepare a Report**:
   Compile the screenshots into a report file. Ensure the report is:
   - Neatly organized with proper headings (e.g., "Instruction Memory Initialization", "Processor Execution", "Verification Results").
   - Includes brief descriptions under each screenshot.
   - Formatted professionally (consistent font, clear spacing, page numbers if needed).
5. **Submission**:
   Attach the completed report file along with the screenshots and submit it as per the given deadline.